

1

Problem

GLOBAL CONTEXT AT A COST

- Self-attention provides global receptive fields but costs $O(N^2)$; CNNs remain local.
- State-space vision models improve complexity but can be difficult to parallelize and interpret.
- The goal is a global operator with strong scalability, hardware efficiency and physical meaning.

4

Mathematics

SPECTRAL SOLUTION

- $du/dt = k (d^2u/dx^2 + d^2u/dy^2)$
- $U_t(w_x, w_y) = \exp[-k(w_x^2 + w_y^2)t] U_0(w_x, w_y)$
- DCT-based evaluation gives a global receptive field at roughly $O(N^{1.5})$, below quadratic attention.

2

Core idea

SEMANTICS AS HEAT

- Treat image patches as heat sources and visual information as thermal energy on a 2D plate.
- Diffusion propagates information globally while naturally suppressing high-frequency components.
- A learned thermal diffusivity makes the amount of smoothing input-dependent.

VHEAT

Building Vision Models upon Heat Conduction

A Physics-Inspired Global Vision Backbone | 2025

Model visual semantics as thermal energy diffusing over a 2D field; solve the heat equation in the frequency domain for global, adaptive and efficient token mixing.

heat equation

frequency

global vision

5

Evidence

ACCURACY + EFFICIENCY

- ImageNet: vHeat-B reaches 84.0% top-1 at 11.2G FLOPs, outperforming Swin-B and Vim-B.
- At 768 resolution it delivers about 3x throughput and 80% lower GPU memory than Swin.
- COCO APbox reaches 47.7 and ADE20K mIoU 49.6; robustness and low-level restoration also improve.

3

HCO

HEAT CONDUCTION OPERATOR

- Transform features with a 2D DCT, multiply frequency coefficients by the heat kernel, then apply IDCT.
- Frequency Value Embeddings predict adaptive diffusivity and are shared within each backbone stage.
- HCO becomes a plug-in global mixer inside hierarchical vHeat-T/S/B models.

6

Meaning & limits

TAKEAWAY

- Heat conduction offers an interpretable frequency-filtering view of global visual modeling.
- The operator scales well to high resolution and transfers across recognition and restoration tasks.
- Its DCT boundary assumptions and frequency prior may not fit every visual domain; FVEs require resizing for new resolutions.